

PURIFICATION:

Nov. 13/25 ①

First we need to cover a new linear operator: $\text{vec}(\cdot)$.

VEC/UNVEC Operations.

- will see this used several times in QIT (and elsewhere).
 - a useful algebraic & computational tool
- it's a linear operator in its own right
 - can be written in an ONB in coord form as a matrix
 - has a guaranteed unique adjoint operation.
 - turns out it is unitary: adjoint = inverse.
- in the most concrete terms:

"It takes a matrix & turns it into a vector by stacking its columns"

• example: $\text{vec} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a \\ c \\ b \\ d \end{pmatrix}$, $\text{unvec} \begin{pmatrix} a \\ c \\ b \\ d \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

→ this is usually how it is introduced

- this ignores the algebraic (abstract) properties
 - dependence of the Hilbert space
 - " on the basis chosen.

① • As a matrix map you can consider $m \times n$ matrices $\bar{c}$ entries in $\mathbb{C}$ with basis matrices E_{ij} .

→ $E_{11} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}_{3 \times 2}$, $E_{21} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \end{pmatrix}_{3 \times 2}$, etc.

• also column vectors indexed by order pairs (i,j) :

$e_{00} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}_{6 \times 1}$ $e_{21} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}_{6 \times 1}$... $e_{32} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}_{6 \times 1}$

- we can write vec as the linear operation:

$$\text{vec}: M_{m \times n}(\mathbb{C}) \longrightarrow \mathbb{C}^{mn}$$

$$A = \sum_{ij} A_{ij} E_{ij} \longmapsto \sum_{ij} A_{ij} e_{(ij)}$$

eg. $\begin{pmatrix} 3 & 1 \\ 5 & 7 \\ 9 & 2 \end{pmatrix} = 3 \cdot \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix} + \dots \longmapsto \begin{pmatrix} 3 \\ 5 \\ 9 \\ 1 \\ 7 \\ 2 \end{pmatrix} = 3 \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \dots$

- unvec just reverses the operation $\rightarrow$ it's a bijection.

- this was just fancy notation to say, "stack the columns".

$\Rightarrow$ now look at A as a linear operator

- start with an $A_{m \times n}$ matrix $\rightarrow$ takes an n -vector outputs an m -vector

$$\begin{bmatrix} A \end{bmatrix}_{m \times n} \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}_{n \times 1} = \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}_{m \times 1}$$

- let H_{in} and H_{out} be Hilbert spaces $\subset$ ONB's: $\{|j\rangle_{\text{in}}\}_{j=1}^n$
- as an operator mapping from, $\{|i\rangle_{\text{out}}\}_{i=1}^m$

$$H_{\text{in}} \longrightarrow H_{\text{out}}$$

we can write it in terms of the basis kets (also operators):

$$A = \sum_{ij} A_{ij} |i\rangle_{\text{out}} \langle j|_{\text{in}}$$

- Note: we extract the (k,l) matrix elts of the operator

$$\text{as usual: } \langle k_{\text{out}} | A | l_{\text{in}} \rangle = \langle k_{\text{out}} | \sum_{ij} A_{ij} |i\rangle_{\text{out}} \langle j|_{\text{in}} | l_{\text{in}} \rangle$$

$$= \sum_{ij} A_{ij} \langle k | i \rangle \cdot \langle j | l \rangle \leftarrow \langle i | \rangle \text{'s are inner products}$$

$$= A_{kl} \text{ as expected.}$$

= δ 's.

(3)

- we can now write the vec operator as:

$$\text{vec}: L(H_{\text{in}}, H_{\text{out}}) \longrightarrow H_{\text{out}} \otimes H_{\text{in}}$$

operator vector space

$$A = \sum_{ij} A_{ij} |i\rangle_{\text{out}} \langle j|_{\text{in}} \longmapsto \text{vec}(A)$$

$$\equiv \sum_{ij} A_{ij} |i\rangle_{\text{out}} \otimes |j\rangle_{\text{in}}$$

NOTE: A linear operator:
 $H_{\text{in}} \rightarrow H_{\text{out}}$

NOTE: A vector
in $H_{\text{out}} \otimes H_{\text{in}}$.

- unvec just reverses the map $|i\rangle \rightarrow$

$$\text{unvec}: H_{\text{out}} \otimes H_{\text{in}} \longrightarrow L(H_{\text{in}}, H_{\text{out}})$$

vector operator.

$\rightarrow |i\rangle_{\text{out}} \otimes |j\rangle_{\text{in}}$ are basis vectors!

Example:

- assume $H_{\text{in}} \cong H_{\text{out}} \cong \mathbb{C}^2$ with standard basis $\{|0\rangle, |1\rangle\}$.

• let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \iff A = a|0\rangle_{\text{out}}\langle 0|_{\text{in}} + b|0\rangle_{\text{out}}\langle 1|_{\text{in}} + c|1\rangle_{\text{out}}\langle 0|_{\text{in}} + d|1\rangle_{\text{out}}\langle 1|_{\text{in}}$
 $\rightarrow$ an operator.

• $\text{vec} A = \begin{pmatrix} a \\ c \\ b \\ d \end{pmatrix}$

$$\iff \text{vec} A = a|0\rangle_{\text{out}} \otimes |0\rangle_{\text{in}} + c|1\rangle_{\text{out}} \otimes |0\rangle_{\text{in}} + b|0\rangle_{\text{out}} \otimes |1\rangle_{\text{in}} + d|1\rangle_{\text{out}} \otimes |1\rangle_{\text{in}}$$

$$\left. \begin{array}{l} a \\ c \\ b \\ d \end{array} \right\} \begin{array}{l} : |00\rangle \\ : |10\rangle \\ : |01\rangle \\ : |11\rangle \end{array}$$

$\rightarrow$ a vector

VEC IDENTITIES - assumes compatible dim's.

④

① • in matrix form we have the Kronecker product

$$A \otimes B = \left(\sum_{ij} a_{ij} E_{ij} \right) \otimes B.$$

$$= \sum_{ij} a_{ij} (E_{ij} \otimes B).$$

$$= \begin{bmatrix} a_{11}B & a_{12}B & \dots \\ a_{21}B & a_{22}B & \dots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

← can show.
- in block form.

$$\textcircled{2} (\text{vec } X, \text{vec } Y)_{\text{vec}} = \text{tr}(X^\dagger Y)$$

↑
standard vector
inner product

↑
operator
inner product

$$\textcircled{3} \text{vec}(AXB) = (B^T \otimes A) \text{vec } X.$$

Special cases:

$$\bullet \text{vec } AX = (I \otimes A) \text{vec } X$$

$$\bullet \text{vec } XB = (B^T \otimes I) \text{vec } X.$$

④ Partial trace

⑤

- let $X, Y : H_R \rightarrow H_A$
- set $|\Psi_X\rangle = |\text{vec} X\rangle \in H_A \otimes H_R$
 $|\Psi_Y\rangle = |\text{vec} Y\rangle \in "$

Claim: $\text{tr}_R (|\text{vec} X\rangle \langle \text{vec} Y|) = XY^\dagger$

Why?

- recall we can write $\text{tr}_R = (I_A \otimes \text{tr}_R)$
 $\uparrow$ ordinary trace operation on R .
- e.g., let $|\phi\rangle = |a\rangle \otimes |r\rangle$
 for some $|a\rangle \in H_A, |r\rangle \in H_R$.

Then $\text{tr}_R |\phi\rangle \langle \phi|$

$$\begin{aligned} &= \text{tr}_R (|a\rangle \otimes |r\rangle) (\langle a| \otimes \langle r|) \\ &= \text{tr}_R |a\rangle \langle a| \otimes |r\rangle \langle r| \\ &= (I_A \otimes \text{tr}_R) (|a\rangle \langle a| \otimes |r\rangle \langle r|) \\ &= |a\rangle \langle a| \otimes \text{tr}_R |r\rangle \langle r| \\ &= |a\rangle \langle a| \otimes \text{tr}_R \langle r|r\rangle \\ &= |a\rangle \langle a| \otimes \text{tr}_R 1_c \\ &= |a\rangle \langle a| \otimes 1_c \\ &= |a\rangle \langle a| \cdot 1_c \end{aligned}$$

In great detailed steps.

$\therefore \boxed{\text{tr}_R |\phi\rangle \langle \phi| = |a\rangle \langle a|}$

So, expand $|\text{vec} X\rangle$ and $\langle \text{vec} Y|$:

$$|\text{vec} X\rangle = \sum_{ai} x_{ai} |a\rangle_A |i\rangle_R$$

$$\langle \text{vec} Y| = \sum_{bj} y_{bj} \langle b|_A \langle j|_R \quad = (|a\rangle_A \otimes |i\rangle_R) (\langle b|_A \otimes \langle j|_R)$$

$$\begin{aligned} \therefore |\text{vec} X\rangle \langle \text{vec} Y| &= \sum_{\substack{ai \\ bj}} x_{ai} y_{bj}^* |a\rangle_A |i\rangle_R \langle b|_A \langle j|_R \\ &= \sum_{\substack{ai \\ bj}} x_{ai} y_{bj}^* |a\rangle_A \langle b|_A \otimes |i\rangle_R \langle j|_R \end{aligned}$$

$$\therefore \text{tr}_R |i\rangle \langle j| = \sum_{\substack{ai \\ bj}} x_{ai} y_{bj}^* |a\rangle_A \langle b|_A \otimes \underbrace{\text{tr} |i\rangle_R \langle j|_R}_{= \delta_{ij}}$$

$$= \sum_{ab} x_{ai} y_{bi}^* |a\rangle \langle b|$$

$$= \sum_{ab} \left(\sum_i x_{ai} y_{bi}^* \right) |a\rangle \langle b|$$

$$= \sum_{ab} \left(\sum_i [X]_{ai} [Y^\dagger]_{ib} \right) |a\rangle \langle b|$$

$$= \sum_{ab} [XY^\dagger]_{ab} |a\rangle \langle b|$$

$\underbrace{\hspace{1cm}}_{= \text{Eck}}$

$$\therefore \text{tr}_R |\text{vec} X\rangle \langle \text{vec} Y| = XY^\dagger$$

Similarly,

$$\text{tr}_A |\text{vec} X\rangle \langle \text{vec} Y| = X^T Y^*$$

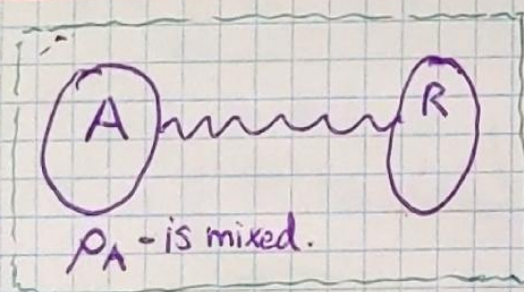
PURIFICATION:

- Basic idea is we may have a quantum system, A , in a general mixed state, ρ_A .
- By adding a new quantum system, R , we can create a pure state, $|\Psi_{AR}\rangle$, in the new larger system such that

$$\text{tr}_R |\Psi_{AR}\rangle \langle \Psi_{AR}| = \rho_A$$

$\Rightarrow |\Psi_{AR}\rangle$ is called the "purification" of ρ_A .

- i.e., the larger system AR as a whole is in a pure state. The isolated system, A , remains mixed as ρ_A .



$$\rho_{AR} = |\Psi_{AR}\rangle \langle \Psi_{AR}| \text{ is pure.}$$

Claim:

Given a density matrix $\rho_A \in L(H_A)$, there exists a $|\Psi_{AR}\rangle \in H_A \otimes H_R$ s.t. $\text{tr}_R |\Psi_{AR}\rangle \langle \Psi_{AR}| = \rho_A$.

Why?

- start w/ the d.m. ρ_A on H_A & take its spectral decomposition.

$$\rho_A = \sum_{i=1}^r \lambda_i |u_i\rangle \langle u_i|, \quad \lambda_i > 0, \quad \sum_i \lambda_i = 1$$

- r is the rank of ρ_A (num. of non-zero eigenvalues).
- form the ONB w/ the eigenvectors, $\{|u_i\rangle\}_{i=1}^r$.
- now define or pick another Hilbert space, H_R w/ $\dim^\infty \geq r$. and an ONB of its own, $\{|k\rangle_R\}_{k=1}^r$.

- take a square root of ρ_A by defining an X such that $\rho_A = XX^\dagger$

• take:
$$X = \sum_{k=1}^r \sqrt{\lambda_k} |u_k\rangle_A \langle k|_R$$

$\uparrow$ H_A $\uparrow$ H_R
 $\nwarrow$ k 'th eigenvector of ρ_A .
 $\swarrow$ k 'th basis vector in R

- NOTE: ① $X: H_R \rightarrow H_A$
 $\therefore X^\dagger: H_A \rightarrow H_R$

$$\Rightarrow X \circ X^\dagger: H_A \xrightarrow{X^\dagger} H_R \xrightarrow{X} H_A$$

So, $\rho_A = XX^\dagger: H_A \rightarrow H_A$ as expected.

- ② This is not the usual square root,

$$X' = \sum_{k=1}^r \sqrt{\lambda_k} |u_k\rangle_A \langle u_k|_A$$

- ③ It still works!

$$\begin{aligned} XX^\dagger &= \left(\sum_k \sqrt{\lambda_k} |u_k\rangle_A \langle k|_R \right) \left(\sum_\ell \sqrt{\lambda_\ell} |u_\ell\rangle_A \langle \ell|_R \right)^\dagger \\ &= \sum_{k\ell} \sqrt{\lambda_k \lambda_\ell} |u_k\rangle_A \langle k|_R (|u_\ell\rangle_A \langle \ell|_R)^\dagger \\ &= \sum_{k\ell} \sqrt{\lambda_k \lambda_\ell} |u_k\rangle_A \underbrace{\langle k|_R |\ell\rangle_A}_{\delta_{k\ell}} \langle u_\ell|_A \\ &= \sum_k \lambda_k |u_k\rangle_A \langle u_k|_A \\ &= \rho_A \quad \checkmark \end{aligned}$$

- Now: Define,

$$|\psi\rangle_{AR} = |\text{vec } X\rangle$$

$$= \sum_{k=1}^r \sqrt{\lambda_k} |u_k\rangle_A \otimes |k\rangle_R$$

using $X = \sum_{k=1}^r \sqrt{\lambda_k} |u_k\rangle_A \langle k|_R$ above.

$\uparrow$ k 'th basis vector of R
 $\nwarrow$ k 'th eigenvector of ρ_A

$\Rightarrow$ This is a Schmidt decomⁿ!

This is the required $|\Psi_{AR}\rangle$.

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Why?

① we constructed $|\Psi_{AR}\rangle = |\text{vec } X\rangle$, where

② we constructed X s.t. $XX^\dagger = \rho_A$.

③ recall the vec-identity:

If $X, Y: H_R \rightarrow H_A$, then

$$\text{tr}_R(|\text{vec } X\rangle\langle\text{vec } Y|) = XY^\dagger$$

④ set $Y = X$.

We then have:

$$\text{tr}_R(|\text{vec } X\rangle\langle\text{vec } X|) = XX^\dagger$$

⑤ We defined $|\Psi_{AR}\rangle$ to equal $|\text{vec } X\rangle$.

$$\begin{aligned} \rightarrow \therefore \text{tr}_R(|\Psi_{AR}\rangle\langle\Psi_{AR}|) &= XX^\dagger \\ &= \rho_A \text{ as required.} \end{aligned}$$

QED.

Note: $|\Psi_{AR}\rangle$ is normalized as it should:

$$\langle\Psi_{AR}|\Psi_{AR}\rangle = \langle\text{vec } X|\text{vec } X\rangle$$

$$= \text{tr } X^\dagger X$$

$$= \text{tr } XX^\dagger$$

$$= \text{tr } \rho_A$$

$$= 1 \checkmark$$